



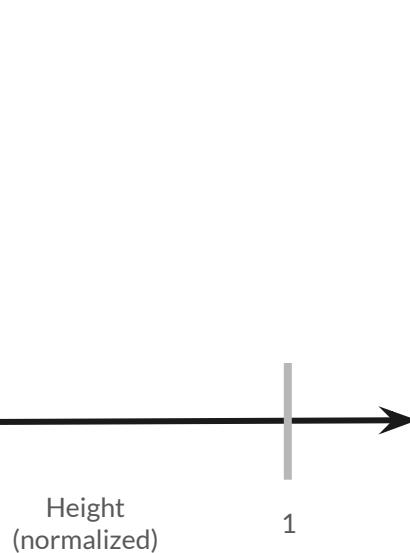
CIS 635 Knowledge Discovery & Data Mining

- Curse of Dimensionality



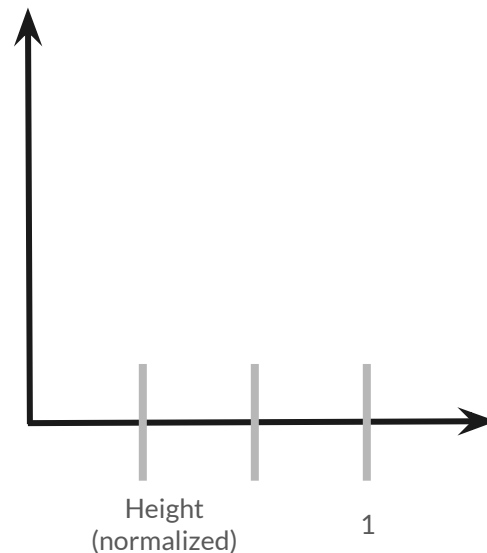
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1])



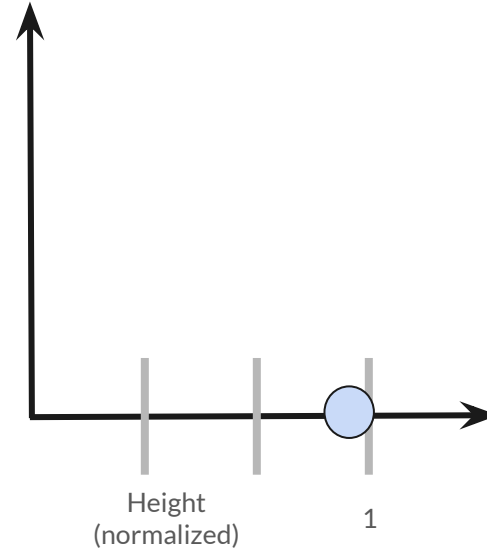
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1])
- 3 equal bins (aka numeric to categorical)



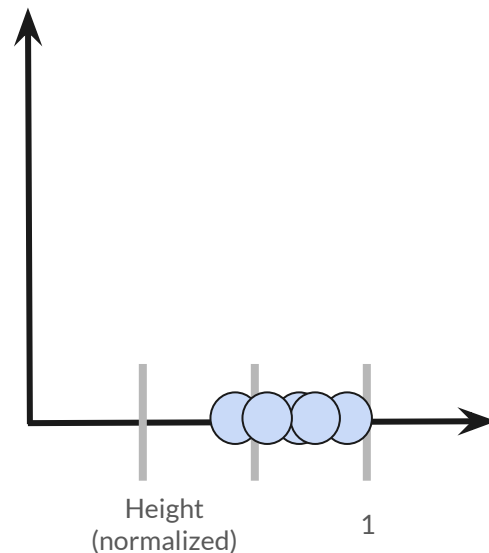
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1])
- 3 equal bins (aka numeric to categorical)
- **The tallest person in the class**



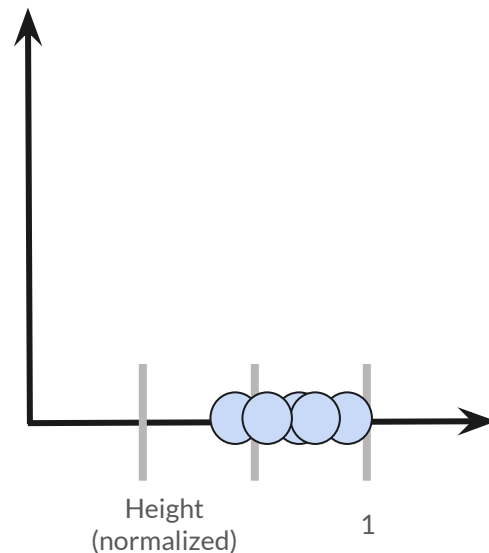
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1])
- 3 equal bins (aka numeric to categorical)
- **All of us**



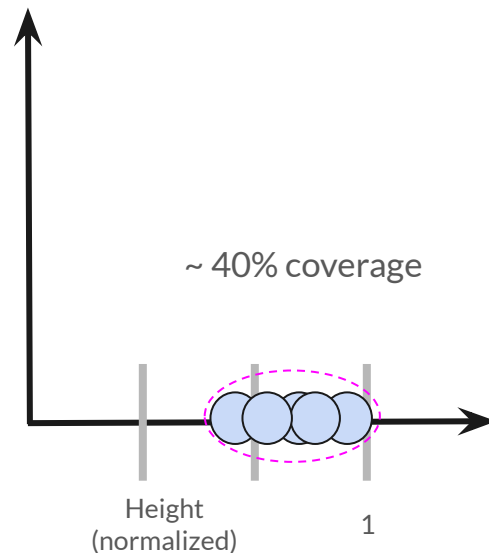
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1])
- 3 equal bins (aka numeric to categorical)
- All of us
- I don't think we have someone with 0.5 times height than the tallest



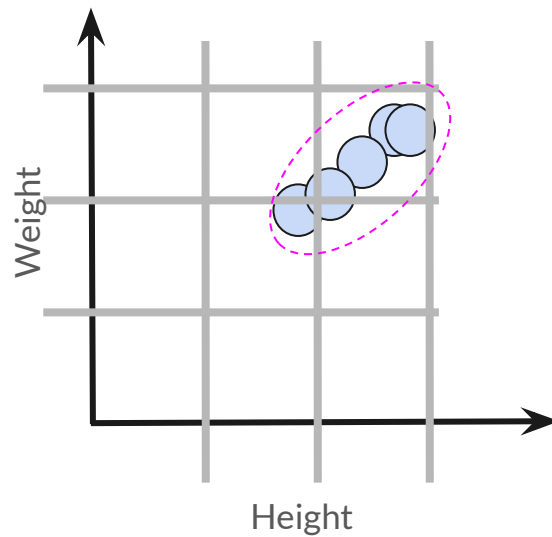
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1])
- 3 equal bins (aka numeric to categorical)
- All of us
- I don't think we have someone with 0.5 times height than the tallest
- Is ~40% domain coverage a reasonable assumption?



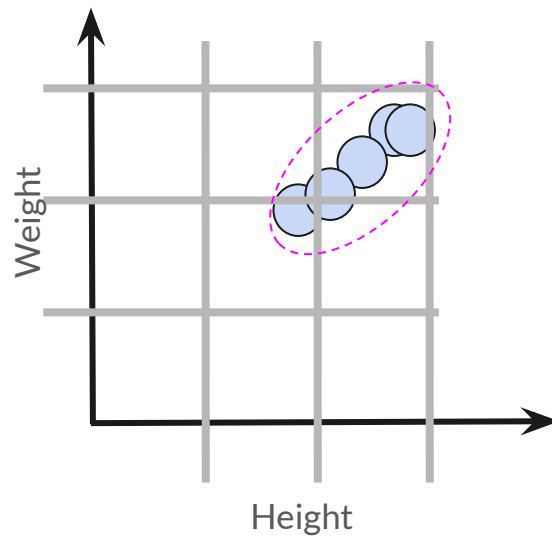
Curse of Dimensionality

- We have added Weight as our second feature dimension
- Do you find the samples represent our class?
- What's the domain coverage now?



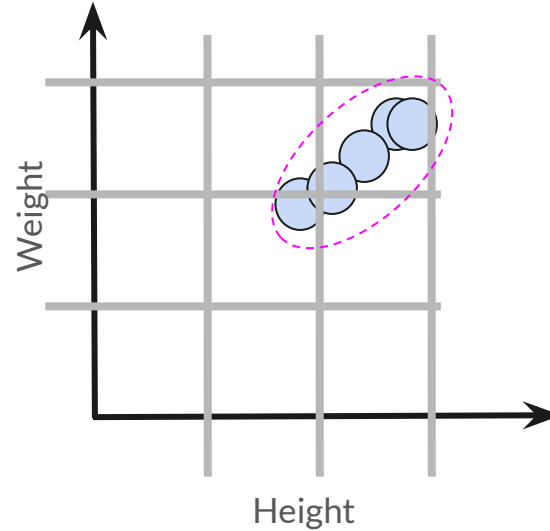
Curse of Dimensionality

- We have added Weight as our second feature dimension
- Do you find the samples represent our class?
- What's the domain coverage now?
- Is ~10% a reasonable assumption?



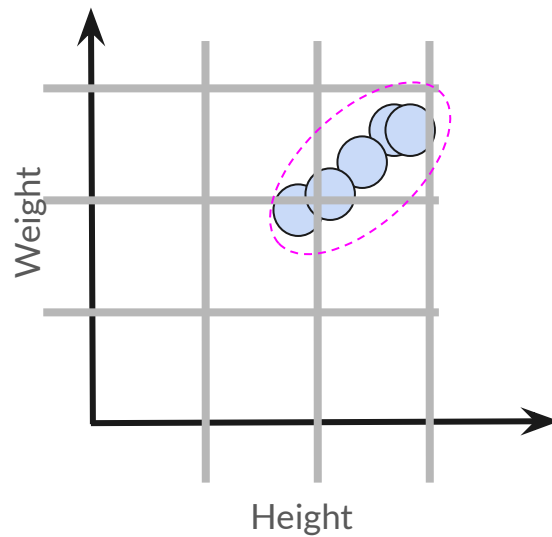
Curse of Dimensionality

- We have added Weight as our second feature dimension
- Do you find the samples represent our class?
- What's the domain coverage now?
- Is ~10% a reasonable assumption?
- **See how numbers dropped from ~40% to ~10%**



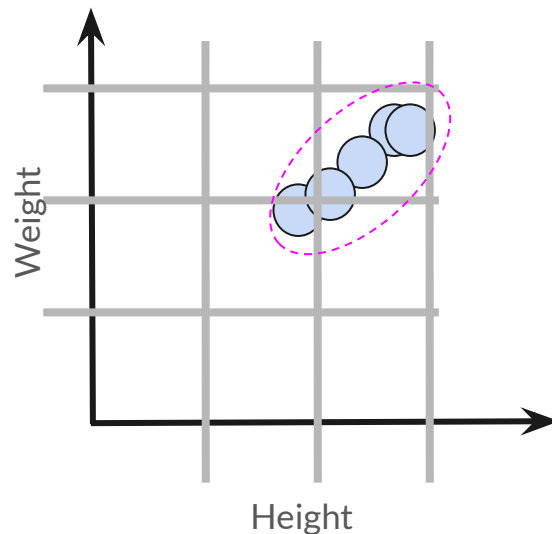
Curse of Dimensionality

- We have added Weight as our second feature dimension
- Do you find the samples represent our class?
- What's the domain coverage now?
- Is ~10% a reasonable assumption?
- **See how numbers dropped from ~40% to ~10%**
- **The other way, the empty space grew up from ~60% to ~90%**



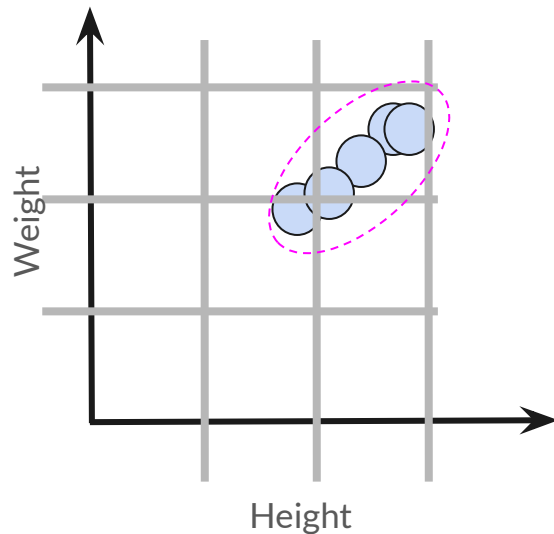
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- What's the domain coverage now?
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- The other way, the empty space grew up from ~60% to ~90%
- What's the number would look like if we add **2** more dimensions?



Curse of Dimensionality

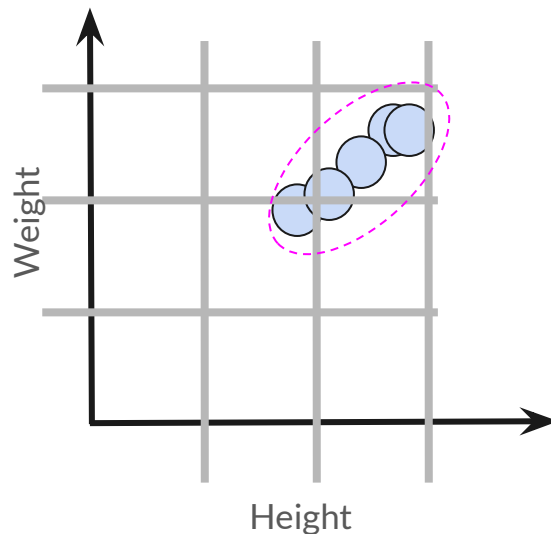
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- Empty space grows exponentially with the increase in adding new features.

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- Empty space grows exponentially with the increase in adding new features.
- Data distribution becomes sparse, and difficult to learn a good model.



Notebook Presentation

Inspecting Pairwise Distances in HD Space



Next Item

- Feature Selection
- Dimensionality Reduction